COMPARATIVE ANALYSIS OF ZERO-SHOT, ONE-SHOT AND FEW-SHOT PROMPTING

Aim

To generate responses for the same task using zero-shot, one-shot, and few-shot prompts, and compare relevance, accuracy, clarity, and consistency.

Procedure

1. Choose a sentiment-classification task for a product review.
2. Create a zero-shot prompt containing only the task instruction.
3. Create a one-shot prompt by adding one labelled example.
4. Create a few-shot prompt with examples of positive, neutral, and negative sentiment.
5. Use the same target review for all methods and record each response.
6. Score the outputs using predefined criteria on a 1–5 scale.
7. Repeat with several reviews to get a more reliable comparison; outputs can vary by LLM and run.

Structured Prompt

```
Task: Classify the review sentiment as Positive, Neutral, or Negative.
Review: "The delivery was quick and the product works perfectly."
Return only one label.

One-shot example:
Review: "The package arrived late and was damaged."
Label: Negative

Few-shot examples:
Review: "Excellent quality; I am very happy." -> Positive
Review: "It is an ordinary product, nothing special." -> Neutral
Review: "The item broke after one day." -> Negative
```

Program / Demonstration

```
review = "The delivery was quick and the product works perfectly."

responses = {
    "Zero-shot": "Positive",
    "One-shot": "Positive",
    "Few-shot": "Positive"
}
for method, answer in responses.items():
    print(f"{method} response: {answer}")

# Illustrative evaluation scores from 1 to 5
scores = {
    "Zero-shot": {"relevance": 5, "accuracy": 5, "clarity": 5, "consistency": 4},
    "One-shot":  {"relevance": 5, "accuracy": 5, "clarity": 5, "consistency": 5},
    "Few-shot":  {"relevance": 5, "accuracy": 5, "clarity": 5, "consistency": 5}
}
print("\\nIllustrative scores:")
for method, score in scores.items():
    print(method, score)
```

Comparison Table

Sample classroom ratings (1 = poor, 5 = excellent). Scores are illustrative rather than measurements from a live LLM.

Method	Prompt details	Relevance	Accuracy	Clarity	Consistency
Zero-shot	Task only	5	5	5	4
One-shot	Task + 1 example	5	5	5	5
Few-shot	Task + 3 examples	5	5	5	5

Google Colab Output Screenshot

Illustrative screenshot only, not captured from a live Colab session. Run the code in Colab for an authentic screenshot if required.

Google Colab Prompting_Comparison.ipynb

```
[1] review = "The delivery was quick..."
    responses = {
      "Zero-shot": "Positive",
      "One-shot": "Positive",
      "Few-shot": "Positive"
    }
    for method, answer in responses.items():
        print(method, answer)
```

Output

```
Zero-shot response: Positive
One-shot response: Positive
Few-shot response: Positive
Illustrative scores (1-5):
Zero-shot consistency: 4
One-shot and few-shot consistency: 5
```

Illustrative Colab-style screenshot; run the code for a live screenshot.

Expected Output

```
Zero-shot response: Positive
One-shot response: Positive
Few-shot response: Positive

Illustrative scores (1-5):
Zero-shot consistency: 4
One-shot consistency: 5
Few-shot consistency: 5
```

Result

For this example, all three methods classify the review as Positive. One-shot and few-shot examples clarify the expected labels and may improve consistency. These scores are illustrative; real evaluation should use several test examples.